Rajkumar Arumugam Jeeva

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WORK EXPERIENCE

Part Time Research Assistant (Computer Vision) Mar 2024 - Present

CHILDREN'S HOSPITAL OF EASTERN ONTARIO | Ottawa

- **Developed** and **optimized computer vision pipelines** using classical and modern algorithms including **pose estimation, object detection, and tracking algorithms**, improving gait video assessment accuracy by **35%**
- **Trained and fine-tuned CNN and Random Forest models** with **TensorFlow** and **scikit-learn**, achieving **95% accuracy** in clinical gait video classification.
- **Automated preprocessing and data quality control** for noisy video inputs, implementing **image denoising, bounding-box filtering, and overlap detection** to enhance input reliability.
- **Deployed models** to **Azure cloud and edge environments** via **Azure Functions and Blob Triggers**, cutting manual video validation time by **80%**.
- **Monitored and maintained deployed services**, applying **continuous testing, version control, and performance tracking** to ensure stable production performance.

Computer Vision Engineer Aug 2021 - Apr 2023

HTIC - INDIAN INSTITUTE OF TECHNOLOGY MADRAS RESEARCH PARK | Chennai, India

- **Designed and deployed real-time object detection and tracking algorithms** using **OpenCV** and **Python**, improving automated monitoring accuracy by **30%**
- **Built AI-powered quality inspection systems** for biomedical devices using **image segmentation, filtering, and OCR**, ensuring **97% detection precision** in production tests.
- **Optimized inference on embedded edge devices** (Raspberry Pi, RP Pico) using **MicroPython**, enabling low-latency analytics across 10+ networked components.
- **Coordinated project deliverables** with cross-functional teams, ensuring timely completion and compliance with research and reporting standards.
- Managed data processing and **version control** using **Git**, ensuring streamlined development and model retraining for optimized deployment.

SKILLS

Programming: Python, C, C++(basic)

Computer Vision: OpenCV, TensorFlow, PyTorch, scikit-image, image denoising, OCR, object detection, tracking

MLOps & Deployment:, Azure Functions, MLFlow, Docker, Git, Edge AI (RPi, MicroPython)

AI on cloud/edge: Azure Functions, Blob triggers, SQL

Core Expertise: AI algorithm design, video analytics, model optimization, cloud/edge deployment, performance testing

Soft Skills: Clear Communication, structured reporting, teamwork

PUBLICATIONS

Automated Video Quality Assessment for Edinburgh Visual Gait Score

MDPI

Designed, implemented and validated a machine learning framework to automate video quality assessment for Edinburgh Visual Gait Score (EVGS)

Parametrization of Optical Flow Kymograms

IEEE

Developed optical flow kymograms and glottal optical flow waveforms, enabling visual assessments and quantitative analysis of vocal fold parameters.

PROJECTS

Classification of Brain Anomalies Using MRI (Python, SQL, FastAPI, Heroku)

- Developed and deployed an **end-to-end ML pipeline** using **MobileNet CNN**, **SQL**, and **FastAPI**, achieving **98.6% accuracy** and **production deployment on Heroku** for real-time inference.

AI-Powered Automatic Cold Mail Generator (Python, Llama 3.1, LangChain, ChromaDB, Streamlit)

- Built an **LLM-based automation pipeline** integrating **semantic search**, **data retrieval**, and **text generation** with **modular API design** for scalable deployment.

Facial and Object Recognition Pipeline (Python, TensorFlow, OpenCV)

- Built an **AI pipeline** integrating **object detection and facial recognition models** for automated monitoring using **TensorFlow and OpenCV**, optimized for real-time video streams

EDUCATION

MASTER of APPLIED SCIENCE | University of Ottawa

Thesis : Automated Objective Video Quality Assessment for EVGS Videos (Gait Videos).

Courses: Pattern Classification and Experimental Design, Data Science for Biomedical Engineers, Biomedical instrumentation

CGPA : 9.2/10